



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT1011
Course title (English)	Honours Calculus I
Course title (Chinese)	微积分荣誉课程 (一)
Units	3
Language of Instruction	English
Description (English)	This course and its continuation MAT1012 are designed for those students who are interested in mathematics, and who wish to build a solid foundation for more advanced courses. These courses are aimed to develop students' abilities of analysing, problem-solving, proving and standardized expression. The course MAT1011 deals with the basic theory and skills of calculus of single variable. Topics include functions, limits and continuity, differentiation, integration and their applications.
Description (Chinese)	本课程教授单变量微积分的基本原理和方法，课程内容包括函数、极限与连续、微分、积分，以及它们的应用。MAT1011及其后续课程MAT1012都是面向对数学有兴趣的学生，为学习后续课程提供坚实的基础，并培养学生分析问题、解决问题、数学论证和规范表达的能力。

2. Prerequisites / Co-requisites

A. Prerequisites

B. Co-requisites

3. Learning Outcomes

Upon successful completion of the courses, the students will be able to:



- Understanding the concept of limit and continuity of functions, and capacity to examine continuity and discontinuity of a function, and analyze properties of continuous functions.
- Understanding derivatives and differentials of functions, and capacity to find derivatives of elementary functions and their inverse functions, verify monotonicity and concavity of a function by using theorems for differentiable functions, find extreme values and sketch curves, and analyze indeterminate forms by using L'Hôpital's rule.
- Understanding the concept of Riemann sums and definite integrals, and capacity to examine integrability or non-integrability of a function. Acquisition of techniques of evaluating definite integrals, including the substitutions, integration by parts, trigonometric substitutions and partial fractions.
- Capacity to find arc lengths, areas of plane regions, areas of surfaces of revolution, volumes, moments and centers of mass.
- Understanding the concept of antiderivatives and indefinite integrals, and the fundamental theorem of calculus.
- Skills to verify convergence or divergence of improper integrals and to evaluate the integrals.
- Knowledge of elementary functions, and capacity to analyze more general functions by using theorems and methods from calculus.
- Skills to solve first order differential equations, and analyze solutions by phase plane method.

4. Course syllabus

- Limit: limit of a function, limit laws, the precise definition of a limit, one-sided limits, sequences and limits
- Continuity: definition of continuity of a function, continuity of elementary functions, intermediate value theorem and extreme value theorem, infinite limits, asymptotes of graphs
- Differentiation: derivative of a function, one sided derivatives, rules of differentiation, differentiation of implicit and inverse functions, derivatives of elementary functions, higher-order derivatives, linearization and differentials, chain rule, implicit differentiation. Extreme and mean value theorem, monotonicity, concavity, the first and second derivative tests, curve sketching, optimization, Newton's method
- Integrals: antiderivatives and indefinite integrals, Riemann sums, definite integrals, integrable and non-integrable functions, rules of definite integrals, areas, mean value theorem for definite integrals, fundamental theorem of calculus.
- Techniques of integration: integral submissions, integration by parts, integrals of rational functions by partial fractions, trigonometric substitutions.
- Applications of definite integrals: volumes via cross sections, arc length, area, work,



moments

- Transcendental functions: properties, derivatives and integrals. Cauchy's mean value theorem, indeterminate forms and L'Hôpital's rule, relative rates of growth, Taylor formula
- Improper integrals: Cauchy criterion, tests for convergence and divergence for improper integrals of non-negative functions and of functions which change sign, Dirichlet test, Abel test, integrands with vertical asymptotes, Cauchy criterion for sequences, monotone convergence theorem
- First order differential equations: basic concepts, slope fields, Euler method. Linear equations, separable equations, applications, introduction to the existence and uniqueness of the first order equations, systems of equations. Autonomous equations, phase-plane analysis

5. Assessment Scheme

Component/ method	% weight
Assignments	20
Quizzes	20
Mid-Term Exam	20
Final Exam	40

6. Grade descriptor

General

Grade	Description
A	• Demonstrate an excellent understanding of the concepts of limits, continuity, differentiation and integration, and a strong ability to clearly and logically apply fundamental theorems and methods of differentiation and integration and theorems of continuous and differentiable functions to solve problems in calculus. Demonstrate the ability of solving unfamiliar problems.
B	• Demonstrate a good understanding of the concepts of limits, continuity, differentiation and integration, and a substantial ability to satisfactorily apply fundamental theorems and methods of differentiation and integration and theorems of continuous and differentiable functions to solve problems in calculus.
C	• Demonstrate an acceptable understanding of the concepts of limits, continuity, differentiation and integration, and an average ability to correctly apply fundamental theorems and methods of differentiation and integration and theorems of continuous and differentiable functions to solve problems in calculus.
D	• Demonstrate a superficial understanding of the concepts of limits, continuity, differentiation and integration, being able to identify the appropriate theorems and



	methods of differentiation and integration and theorems of continuous and differentiable functions when solving a calculus problem, but with substantial inadequacies in applying the theorems or rules.
F	• Demonstrate a poor understanding of the course materials. • Non-ability in applying some basic methods to solve simple problems.

7. Feedback for evaluation

- Formal course and teaching evaluation
- Informal feedback and/or teaching assistant
- School retreat and programme review

8. Reading

Required Reading

1.13, England, Thomas' Calculus (13th Edition), George B. Thomas Jr., Joel Hass, Christopher Heil, Maurice D. Weir.
Publisher: Pearson.

9. Course components

Activity	Hours/week
Lecture	5
Tutorial	2

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Limit of a function, limit laws, the precise definition of a limit, one-sided limits,
00000 00002	Definition of continuity, properties of continuous functions, intermediate value theorem and extreme value theorem, infinite limits, asymptotes of graphs, sequences and limits
00000 00003	Definition of derivatives, calculating derivatives, one sided derivatives, differentiation rules, derivative as a function, second and higher order derivatives, analysis meaning of derivatives, chain rule, implicit differentiation



00000 00004	Linearization and differentials, differentiability and continuous differentiability, extreme values of functions, mean value theorem, monotonicity, first derivative test
00000 00005	Concavity, second derivative test, curve sketching, optimization, Newton's method. Antiderivatives and indefinite integrals, Riemann sums, definition of definite integrals
00000 00006	Integrable and non-integrable functions, rules of definite integrals, area under a graph, mean value theorem for definite integrals
00000 00007	The fundamental theorem of calculus, integral substitution method, integration by parts, applications of integrals: area between curves
00000 00008	Volume using cross-sections, arc length, areas of surfaces of revolution, applications of integrals in physics: work, fluid forces, moments, centers of masses
00000 00009	Inverse functions and their derivatives, natural logarithms and exponential functions, their derivatives and integrals, the number e and related limits
00000 00010	Inverse trigonometric functions and hyperbolic functions, Cauchy's mean value theorem, indefinite forms and L'Hôpital's rule
00000 00011	Relative rates of growth, Taylor formula. Techniques of integration: integration using basic formulas, trigonometric integrals, trigonometric substitutions
00000 00012	Integration of rational functions by partial fractions, reducing to integrals of rational functions, concepts of improper integrals
00000 00013	Cauchy criterion for improper integrals, tests for convergence and divergence for improper integrals of non-negative functions and of functions which change sign, Dirichlet test, Abel test, integrands with vertical asymptotes, Cauchy criterion for sequences, monotone convergence theorem. First order ODE: basic concepts, slope fields, Euler method
00000 00014	Linear equations, separable equations, applications, introduction to the existence and uniqueness of the first order equations, systems of equations. Autonomous equations, phase-plane analysis

11. Any other information

This course is a substitution of MAT1001 Calculus I. The students who intend to declare for Mathematics and Applied Mathematics are strongly recommended to take this one instead of MAT1001.

12. Version date

Version number	2
As of (date)	2023/08/03